

Software Requirements Specification

Capstone Project Requirements Review Desktop Application

Version: 1.0

Platform: Windows / macOS / Linux Desktop

Technology: Flutter Desktop

Primary purpose: Import, analyze, review and evaluate the Requirements section of a Capstone Project document.

Document status: MVP SRS

1. Introduction

1.1 Purpose

The application allows a Capstone Project Reviewer to import a project document, extract its Requirements section, analyze requirement quality with AI, and perform manual review.

- Detect unclear or ambiguous requirements.
- Identify incomplete requirements.
- Evaluate testability and feasibility.
- Detect potential inconsistencies and duplicates.
- Provide reviewer comments, status and an exportable review report.

1.2 Scope

In Scope

- Import and drag-and-drop PDF, DOCX, TXT and Markdown files.
- Document parsing and Requirements section detection.
- Requirement extraction and ID generation.
- Requirement list, detail view, search, filter and sort.
- Manual review and reviewer comments.
- AI analysis of requirement quality.
- AI score, issues, severity and suggested revision.
- Review dashboard and PDF/CSV report export.

Out of Scope

- Multi-user collaboration and real-time editing.
- Online document editing and cloud storage.
- User authentication.
- Complex version control.
- General-purpose AI chatbot.
- Virtual 3D functionality.
- Custom AI model training.
- Mobile application.

2. User Roles

2.1 Reviewer

- Import documents.
- View extracted requirements.
- Run AI analysis.
- Review AI findings.
- Set requirement status.
- Add and edit comments.
- Export the review report.

3. Functional Requirements

ID	Feature	Requirement
FR-001	Open Document	User can open a document from the local computer. The application shall provide a file picker, validate the selected file, process valid files and display an error for invalid files.
FR-002	Drag and Drop	User can drag a supported document into the application. The drop zone shall provide visual feedback while a file is being dragged.
FR-003	File Validation	The application shall validate file extension, size, readability and basic structure. Supported MVP formats are PDF, DOCX, TXT and MD.
FR-004	Extract Document Content	The application shall extract text from supported documents and normalize it for downstream processing.
FR-005	Document Structure Detection	The application shall attempt to detect sections such as Functional Requirements and Non-functional Requirements. If Requirements cannot be detected, the user shall be able to select a section manually.
FR-006	Extract Requirements	The application shall identify individual requirements from the selected Requirements content.
FR-007	Requirement ID	Existing requirement IDs shall be preserved. If IDs are absent, the application shall generate sequential IDs such as REQ-001.
FR-008	Requirement Type	Requirements may be classified as Functional, Non-functional, Business, Technical, Security, Performance, Usability or Other.
FR-009	Requirement Detail	The requirement detail view shall show ID, title, description, type, source location, AI score, review status and reviewer comments.
FR-010	Manual Review Status	The reviewer shall be able to set a requirement to Not Reviewed, Passed, Needs Review or Failed.
FR-011	Reviewer Comment	The reviewer shall be able to add, edit and delete comments associated with a requirement.
FR-012	Analyze Requirement	The reviewer shall be able to submit a requirement to the AI service for structured quality analysis.
FR-013	AI Evaluation Criteria	AI analysis shall evaluate Clarity, Completeness, Testability, Consistency, Feasibility, Ambiguity and Duplication.
FR-014	AI Result	AI output shall contain a requirement ID, overall score, status, individual quality scores, issues, severity and suggested revision.
FR-015	AI Issue Severity	Detected issues shall be categorized as Low, Medium or High severity.
FR-016	Suggested Revision	The AI may propose a revised requirement. The original requirement shall not be overwritten automatically; the reviewer must explicitly approve a replacement.
FR-017	Batch AI Analysis	The reviewer shall be able to analyze multiple requirements. Progress, completed items and retryable failures shall be visible.
FR-018	AI Provider Fallback	The application shall use an AI provider abstraction so that requests can be routed between available providers when a provider is unavailable or rate-limited.
FR-019	AI Failure Handling	Provider failures, invalid structured responses and rate limits shall result in recoverable UI states and retry/fallback behavior rather than application crashes.
FR-020	Export Review Report	The application shall export a review report containing project information, overall score, review statistics, requirement-level results and identified issues. PDF is the primary MVP export; CSV is optional.

4. AI Analysis Specification

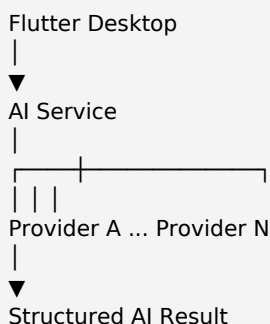
4.1 Evaluation Criteria

Criterion	Evaluation
Clarity	Detect vague language, undefined terms and multiple interpretations.
Completeness	Check whether actor, action, behavior, conditions, constraints and expected result are sufficiently specified.
Testability	Determine whether objective test cases or acceptance conditions can be derived.
Consistency	Compare requirements for conflicting rules or incompatible statements.
Feasibility	Assess whether the requirement appears implementable using the technical context available in the document. The AI must avoid absolute claims when context is insufficient.
Ambiguity	Identify context-dependent vague terms such as 'quickly', 'easy' or 'user-friendly' when they make the requirement difficult to verify.
Duplication	Identify requirements that substantially overlap with other requirements.

4.2 AI Response Contract

```
{
  "requirementId": "REQ-003",
  "overallScore": 68,
  "status": "needs_review",
  "scores": {
    "clarity": 80,
    "completeness": 70,
    "testability": 45,
    "consistency": 90,
    "feasibility": 70
  },
  "issues": [
    {
      "type": "ambiguity",
      "severity": "medium",
      "description": "The term 'quickly' is not measurable."
    }
  ],
  "suggestedRevision": "The system shall..."
}
```

4.3 AI Provider Architecture



The Flutter UI shall depend on an AI service abstraction rather than individual provider implementations. Provider selection, rate-limit handling, cooldown and retry behavior should be encapsulated below the UI layer.

For a desktop MVP, API credentials should not be treated as secure secrets if embedded in the application package. A backend/proxy is the safer architecture when provider credentials must remain

confidential.

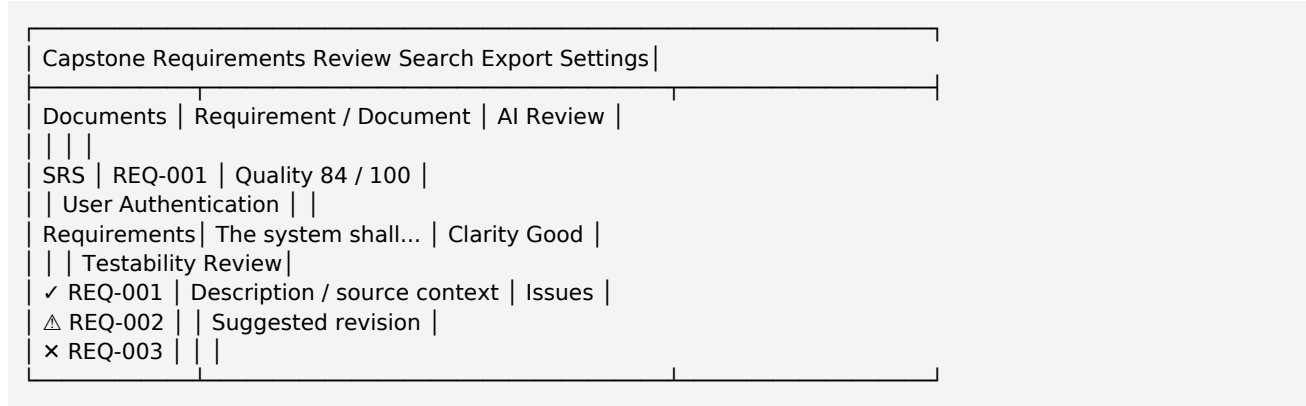
5. User Interface and UX Requirements

5.1 Design Direction

Style: Professional Document Review / Developer Tool, combining editorial document reading patterns with IDE and enterprise productivity conventions.

- Neutral-first visual system.
- No excessive gradients.
- No glassmorphism.
- No decorative AI illustrations.
- No excessive rounded cards.
- No floating chatbot as the primary interaction model.
- Functional cards only.
- Subtle, purposeful animations.

5.2 Main Workspace



Recommended desktop proportions: left sidebar approximately 220–240 px; main content flexible; AI panel approximately 320–380 px.

5.3 Typography and Iconography

- Primary UI font: Inter.
- Optional code font: JetBrains Mono.
- Icon set: Lucide Icons.
- Important actions should use icon + text rather than icon-only controls.
- Icon-only controls require tooltips.
- Suggested typography: page title 20–24 px, section title 15–16 px, body 14 px, secondary text 13 px, captions 12 px.

5.4 Color and Component Rules

- Primary background: neutral off-white/light gray.
- Primary text: near-black.
- Borders: subtle gray.
- Use one primary brand/action color rather than multiple decorative colors.

- Green represents Passed, amber represents Needs Review, red represents Failed and gray represents Not Reviewed.
- Use 1 px borders and minimal shadows.
- Recommended corner radius: approximately 6–10 px.
- Use spacing tokens based on 4/8/12/16/24/32 px.

5.5 Requirement List

REQUIREMENTS 24

Search requirements...

[All] [Needs Review] [Failed]

✓ REQ-001 User Registration
 ✓ REQ-002 User Login
 △ REQ-003 Password Recovery
 2 issues
 × REQ-004 Payment Processing
 1 critical issue

5.6 AI Review Panel

AI REVIEW

Overall
 84 / 100

QUALITY
 Clarity Good
 Testability Needs review
 Completeness Good

ISSUES
 △ Ambiguity
 "quickly" is not measurable.

SUGGESTED REVISION
 The system shall...

[Apply suggestion]

AI should behave as a structured review analyst, not as a conversational assistant. The reviewer should remain in control of accepting any suggested revision.

6. Dashboard and Reporting

6.1 Review Dashboard

CAPSTONE REQUIREMENTS REVIEW

Document
 Capstone_SRS.pdf

24 Requirements

18 Passed 4 Needs Review 2 Failed

Overall Quality
 82 / 100

Review Progress

75%

Recent Issues

REQ-003 Ambiguous terminology
REQ-008 Missing acceptance criteria
REQ-014 Possible duplicate

6.2 Export Report

- Project and document information.
- Review date.
- Overall score.
- Requirement counts by status.
- Requirement-level score and status.
- Issues and severity.
- Suggested revisions.
- Reviewer comments.

7. Search, Filter and Navigation

- Search by requirement ID, title or text.
- Filter by All, Not Reviewed, Passed, Needs Review and Failed.
- Filter by issue severity.
- Keyboard navigation between requirements.
- Selected requirement remains synchronized with the document and AI review panels.

8. Keyboard Shortcuts

Shortcut	Action
Ctrl/Cmd + O	Open document
Ctrl/Cmd + S	Save review
Ctrl/Cmd + F	Search
Ctrl/Cmd + E	Export
Ctrl/Cmd + Enter	Analyze selected requirement
Esc	Close panel/dialog
↑ / ↓	Navigate requirements

9. Accessibility

- Keyboard navigation shall be supported.
- Focus states shall remain visible.
- Text contrast shall be sufficiently high.
- Color shall not be the only indicator of status.
- Icon-only buttons shall have tooltips.
- Interactive controls should be large enough for reliable mouse interaction.

10. Non-Functional Requirements

NFR-001 Performance

Document parsing, AI analysis and export must not freeze the UI. Heavy operations should run asynchronously/background.

NFR-002 Reliability

AI provider failure must be recoverable through retry, fallback or a clear error state. The application must not crash because of a provider response failure.

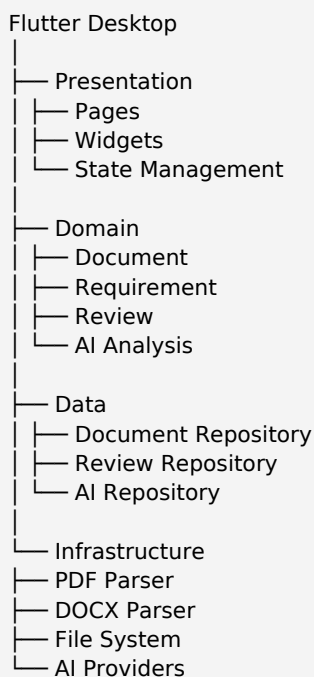
NFR-003 Security

API credentials shall not be hard-coded in source control. Imported documents shall only be processed after explicit user action. Only the data required for AI analysis should be sent to an external AI provider.

NFR-004 Maintainability

AI providers and document parsers shall be abstracted behind application interfaces/repositories so that implementations can be replaced without rewriting the UI.

11. Recommended Architecture



12. Core Data Models

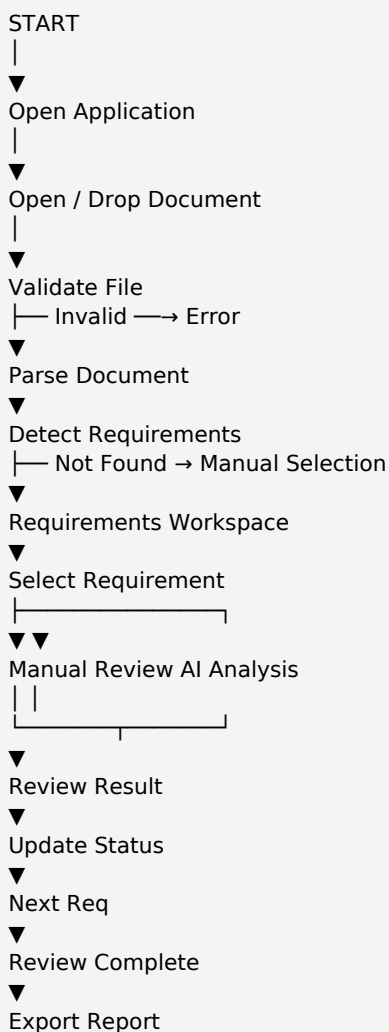


- └ description
- └ type
- └ sourceLocation
- └ status
- └ review
- └ comments[]

Review

- └ overallScore
- └ clarity
- └ completeness
- └ testability
- └ consistency
- └ feasibility
- └ issues[]
- └ suggestedRevision

13. Main User Flow



14. MVP Definition

The MVP is complete when a reviewer can take a Capstone SRS document, import it into the desktop application, identify the Requirements section, extract individual requirements, review each requirement manually and with AI, inspect structured AI findings, set status and comments, and export a review report.

- Phase 1: Flutter Desktop + File Handling + Document Parsing.

- Phase 2: Requirement Extraction + Requirements Workspace + Manual Review.
- Phase 3: AI Analysis + Structured AI Results + AI Suggestions.
- Phase 4: Dashboard + Export + Error Handling + UX Polish.

15. Definition of Done

- Supported files can be imported reliably.
- Requirements can be extracted or manually selected.
- Each requirement can be opened and reviewed.
- AI analysis produces a structured result matching the application schema.
- AI provider failures are recoverable.
- Original requirement text is never overwritten automatically.
- Reviewer status and comments are persisted for the current review session.
- The complete review can be exported.
- The desktop UI remains usable during parsing and AI requests.
- The interface follows the defined professional document-review visual system.